

3. Use-Case Flow Specification

Use Case: Reserve Equipment Slot

Actor : Student

Preconditions:

- 1. The student is authenticated in the system.**
- 2. The required equipment exists in the system.**
- 3. The equipment has valid calibration status.**
- 4. The requested reservation slot is within the allowed booking period.**

Postconditions:

- 1. The reservation is successfully stored in the system.**
- 2. The selected equipment slot is locked.**
- 3. A reservation confirmation/token is generated.**
- 4. The equipment cannot be double-booked for the reserved slot.**

Main Success Scenario:

- 1. Student logs into the system.**
- 2. Student views the available lab equipment.**
- 3. Student selects the required equipment.**
- 4. System checks the equipment's calibration status.**
- 5. System displays available reservation slots.**
- 6. Student selects a suitable slot.**
- 7. System verifies that the selected slot is available.**
- 8. Student confirms the reservation.**
- 9. System creates the reservation and locks the selected slot.**
- 10. System generates a reservation confirmation/token.**

Alternate Flow: Equipment Slot Already Reserved:

- 1. Student selects a reservation slot.**
- 2. System detects that the selected slot has already been reserved.**
- 3. System rejects the reservation request.**
- 4. System displays other available slots.**
- 5. Student selects another available slot and continues with the reservation.**